

Intermediate Elective 2

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Set Up

Packages

```
library(tidyverse)
library(readxl)
library(ggtext)
```

```
library(tidyverse)
library(readxl)
library(ggtext)
library(here)
```

```
aquatic_data <- read_excel("../data/aquatic_data.xlsx")
taxon_list <- read_csv("../data/taxon_list.csv")
```

#Problems: 1. Explain Your Inspiration I chose a South China Morning Post-style timeseries line chart, inspired by Steven Ponce's "Coal falls. Renewables rise." visualization, which tracks how proportions of different categories change over time on a dark background with colored lines.

The Aquatic Insects sheet spans 2020–2025 with counts of invertebrate taxa across multiple sites, making a proportion timeseries ideal for showing how the relative community composition shifts year over year.

Year is mapped to the x-axis, proportion of total count to the y-axis, and taxon group (Copepod, Ostracod, Cladocera, Amphipod, Chironomid) to line color.

2. Plan Your Figure ## Sketch

![My figure sketIMG_6449.jpeg)

```
insects <- read_excel("../data/aquatic_data.xlsx",  
                      sheet = "Aquatic Insects")
```

```
aquatic_data <- read_excel(here("data", "aquatic_data.xlsx"))  
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```
insects <- read_excel(here("data", "aquatic_data.xlsx"), sheet = "Aquatic Insects")  
names(insects)
```

```
[1] "Date data entered"  
[2] "Person Entering Data"  
[3] "Date on Vial"  
[4] "Person that sorted the sample"  
[5] "Site"  
[6] "Partial_sample_note"  
[7] "Sample Type"  
[8] "Ostracod"  
[9] "Copepod"  
[10] "Hemiptera: Corixidae (boatman)"  
[11] "Cladocera"  
[12] "Nematode"  
[13] "Diptera - TOTAL"  
[14] "Dipteran sp."  
[15] "Diptera: Chironomid"  
[16] "Diptera: Ceratopogonidae"  
[17] "Diptera : Ceratopogonidae pupa"  
[18] "Diptera: Culicidae"  
[19] "Diptera: Culicidae Larvae"  
[20] "Diptera : Anthomyiidae"  
[21] "Diptera: Ephydriidae"  
[22] "Diptera: Stratiomyidae"  
[23] "Amphipod"  
[24] "Lepidoptera-Crambidae"  
[25] "Hemiptera TOTAL"  
[26] "Hemiptera- Mesoveliidae"  
[27] "Hemiptera- Notonectidae"  
[28] "Annelida TOTAL"  
[29] "Annelida Sp."  
[30] "Annelida: Oligochaete"  
[31] "Annelida: Polychaete"
```

```

[32] "Ephemeroptera"
[33] "Odonate"
[34] "Coleoptera - TOTAL"
[35] "Coleoptera sp."
[36] "Coleoptera: Dytiscidae"
[37] "Coleoptera: Dytiscidae: Liodes affinis"
[38] "Coleoptera: Dytiscidae: Agabus"
[39] "Coleoptera: Dytiscidae: Colymbetinae"
[40] "Coleoptera: Curculionidae (Weevil)"
[41] "Coleoptera - Hydrophilidae"
[42] "Coleoptera - Tropiscernus"
[43] "Coleoptera Hydraenidae"
[44] "Coleoptera- Gyrinidae- Gyrinini"
[45] "Ephemeroptera Sp."
[46] "Tipulidae Larva"
[47] "Plecoptera Sp."
[48] "Arachnida: Acari (mite)"
[49] "Trichoptera Sp."
[50] "Collembola (Springtail)"
[51] "Gastropod - snail"
[52] "Mollusk"
[53] "Syrphidae"
[54] "Terrestrials"
[55] "Isopoda, sphaeroma"
[56] "Clam Shrimp"
[57] "Eristalis tenax larvae"
[58] "Larvae"
[59] "Crawfish"
[60] "Fish"
[61] "NZ Mudsnailed"
[62] "Unknown"
[63] "Total #"
[64] "Comments"

```

```

# Read the Aquatic Insects sheet from the Excel file
insects <- read_excel(here("data", "aquatic_data.xlsx"),
                      sheet = "Aquatic Insects")

# Define the five key taxa we want to visualize
key_taxa <- c("Copepod", "Ostracod", "Cladocera",
              "Amphipod", "Diptera: Chironomid")

```

```

# Wrangle data: pivot to long format and calculate
# proportion of each taxon per year
plot_data <- insects %>%
  select(`Date on Vial`, all_of(key_taxa)) %>%
  rename(Date = `Date on Vial`) %>%
  filter(!is.na(Date)) %>%
  mutate(Date = as.Date(Date), Year = year(Date)) %>%
  pivot_longer(cols = all_of(key_taxa),
               names_to = "taxon", values_to = "count") %>%
  mutate(count = replace_na(count, 0),
         taxon = str_trim(taxon),
         taxon = if_else(taxon == "Diptera: Chironomid",
                        "Chironomid", taxon)) %>%
  group_by(Year, taxon) %>%
  summarise(total = sum(count, na.rm = TRUE), .groups = "drop") %>%
  group_by(Year) %>%
  mutate(proportion = total / sum(total) * 100) %>%
  ungroup() %>%
  filter(Year >= 2020, Year <= 2025)

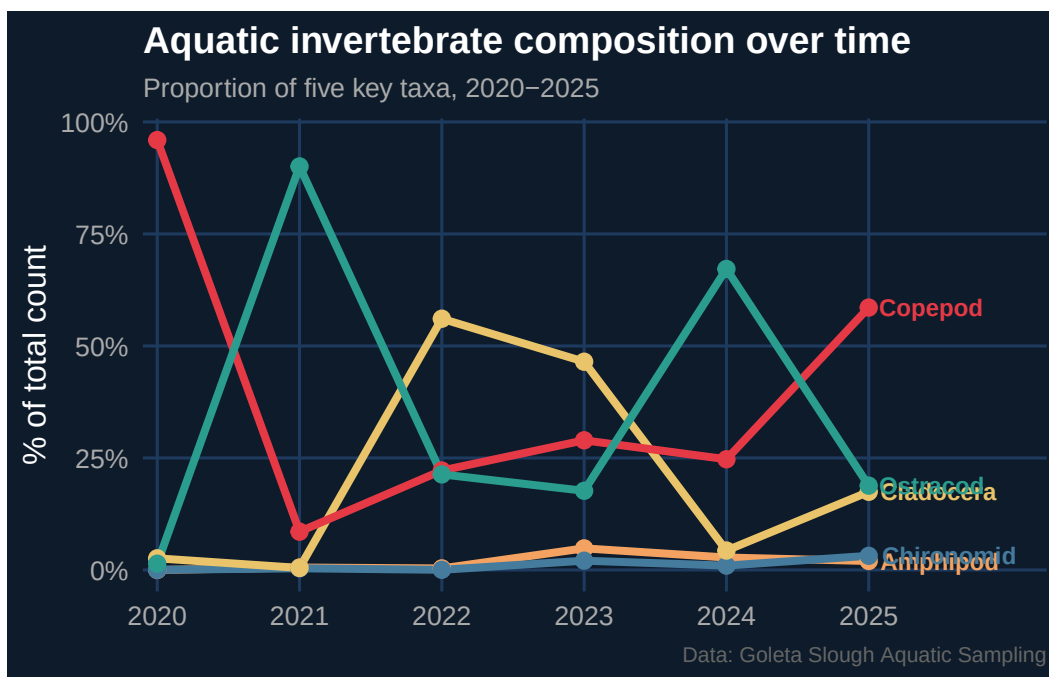
# Define colors for each taxon
colors <- c("Copepod" = "#E63946", "Ostracod" = "#2A9D8F",
           "Cladocera" = "#E9C46A", "Amphipod" = "#F4A261",
           "Chironomid" = "#457B9D")

# Get the last year's values for direct line labeling
labels <- plot_data %>% filter(Year == max(Year))

# Build the plot
ggplot(plot_data, aes(x = Year, y = proportion, color = taxon)) +
  # Add lines and points for each taxon
  geom_line(linewidth = 1.4) +
  geom_point(size = 2.5) +
  # Add direct labels at end of each line instead of a legend
  geom_text(data = labels, aes(label = taxon),
           hjust = -0.1, size = 3.2, fontface = "bold") +
  # Apply custom colors
  scale_color_manual(values = colors) +
  # Format x and y axes
  scale_x_continuous(breaks = 2020:2025,
                    expand = expansion(mult = c(0.02, 0.25))) +
  scale_y_continuous(labels = scales::label_number(suffix = "%")) +

```

```
# Add titles and labels
labs(title = "**Aquatic invertebrate composition over time**",
      subtitle = "Proportion of five key taxa, 2020-2025",
      x = NULL, y = "% of total count",
      caption = "Data: Goleta Slough Aquatic Sampling") +
# Apply dark theme styling
theme_minimal(base_size = 12) +
theme(
  plot.background = element_rect(fill = "#0D1B2A", color = NA),
  panel.background = element_rect(fill = "#0D1B2A", color = NA),
  text = element_text(color = "white"),
  plot.title = element_markdown(size = 14, color = "white"),
  plot.subtitle = element_text(color = "#AAAAAA", size = 10),
  plot.caption = element_text(color = "#666666", size = 8),
  axis.text = element_text(color = "#AAAAAA"),
  panel.grid.major = element_line(color = "#1E3A5F"),
  panel.grid.minor = element_blank(),
  legend.position = "none"
)
```



```
# Save the final figure as a png file
ggsave(here("code", "aquatic_viz.png"), width = 8, height = 5, dpi = 300)
```

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